Notes: Murray, Xia, and Xiao (JFE, 2024)

Charting by Machines

Contents

1 Big picture **2**

2 Data and measurement **3**
- 2.1 Stock universe and periods . . . 3
- 2.2 Dependent variable: excess return and transformations . . . 3
- 2.3 Input variables: cumulative returns in a one-year chart . . . 4
- 2.4 ML-based forecast . . . 4
- 2.5 Optimization design . . . 4
- 2.6 Expanding-window forecasts . . . 4

3 Models and empirical specifications **5**
- 3.1 Portfolio sorting test . . . 5
- 3.2 Risk-adjusted regressions . . . 5
- 3.3 Subperiod and large-stock tests . . . 5
- 3.4 Forecasting-function stability . . . 6
- 3.5 Fama–MacBeth tests for nonlinearities and interactions . . . 6
- 3.6 Momentum and reversal controls . . . 6
- 3.7 Technical-signal and image-based ML controls . . . 6
- 3.8 Optimization-performance validation . . . 7

4 Identification and empirical design **7**
- 4.1 What identifies predictive power . . . 7
- 4.2 Why the result challenges weak-form efficiency . . . 7
- 4.3 Why the design is not just standard anomaly mining . . . 7
- 4.4 What remains imperfect . . . 8

5 Tables and figures **9**
- 5.1 Figure 1 and Table 1: chart information and model optimization . . . 9
- 5.2 Table 2 and Table 3: forecast distribution and main portfolio return result . . . 9
- 5.3 Table 4, Table 5, and Figure 2: risk adjustment and time variation . . . 10
- 5.4 Tables 6–9: large stocks and stability of the forecasting relation . . . 11
- 5.5 Figures 3–4 and Table 10: nonlinearities and interactions . . . 12
- 5.6 Tables 11–12: momentum and reversal . . . 14
- 5.7 Tables 13–16: image-based ML forecasts and technical-signal controls . . . 15
- 5.8 Figures 5–7 and Table 17: chart interpretation and optimization validation . . . 16

6 Short summary **17**

7 Conclusion **18**
- 7.1 Core conclusion . . . 18
- 7.2 Economic intuition . . . 18
- 7.3 Takeaway . . . 18

1 Big picture

- **Question.** Does technical analysis have real predictive power once the mapping from historical price charts to expected returns is learned by machine learning rather than by hand-crafted chart rules?
- **Main empirical test.** The authors test the weak-form efficient market hypothesis by asking whether machine-learning forecasts based only on historical stock performance predict the cross-section of future stock returns.
- **Input information.** The forecast uses only the past 12 monthly cumulative returns, denoted $\mathrm{CR}_1, \ldots, \mathrm{CR}_{12}$. This is information that a chart reader could see in a one-year price plot.
- **Model selection discipline.** The ML architecture and implementation choices are selected only using the 1927:01–1963:06 optimization period. Main return tests use the 1963:07–2022:12 test period.
- **Chosen model.** The optimized model is a convolutional neural network with long short-term memory (CNNLSTM), mean-squared-error loss, equal weight per month and equal weight per stock within month, and normalized excess return as the dependent variable.
- **Main portfolio result.** Value-weighted portfolios sorted on MLER generate a strong return spread. The decile 10-minus-decile 1 portfolio earns about 1.08% per month with a Newey–West t-statistic of 5.51.
- **Risk adjustment.** The return spread remains large under CAPM, Fama–French, Carhart, and models augmented with reversal. Volatility, skewness, value-at-risk, and expected shortfall do not support a risk explanation.
- **Robustness through time.** The signal works in most subperiods, including 2015:01–2022:12, and works among the largest 500 stocks.
- **Stability of learning.** Forecasting functions estimated using different historical windows have correlated signals, overlapping long-short holdings, and correlated long-short returns.
- **Nonlinearity.** The learned function is not just momentum plus reversal. Partial-dependence plots and Fama–MacBeth regressions show important nonlinearities and interactions.
- **Distinctness.** The signal is not subsumed by momentum, reversal, image-based ML forecasts, 14 technical signals from Neely et al., or 14 trading-friction variables from Freyberger et al.
- **Optimization validation.** ML models that perform well in the optimization period continue to perform well out of sample. This supports the authors' ex-ante model-selection procedure.
- **Takeaway.** Price charts appear to contain information that is hard to capture with simple technical rules but exploitable by flexible machine learning. The weak-form EMH is challenged in a highly out-of-sample setting.

Economic roadmap

1. The paper starts from the idea of technical analysis: historical prices are usually viewed as too simple or too noisy to generate reliable abnormal returns.
2. Instead of imposing a small number of human-designed rules such as momentum, reversal, moving-average crossovers, or chart patterns, the authors let a flexible neural network learn the mapping from a one-year price chart to the next month's return rank.
3. The machine-learning model sees only cumulative returns from the chart. It does not see firm fundamentals, accounting variables, analyst forecasts, news, characteristics, liquidity, or factor returns.
4. This restriction is important because it makes the test directly about weak-form efficiency. If the forecast works, the information is contained in past returns alone.

5. The research design separates model choice from model evaluation. The architecture is selected in the early historical sample, while the main asset-pricing tests are conducted in a later test period.
6. The empirical object is not one fitted neural network. The forecast is an average over 30 re-fits, which reduces sensitivity to random initialization and training noise.
7. The core asset-pricing test is a monthly cross-sectional sort. High-MLER stocks are those whose charts imply high future normalized excess returns; low-MLER stocks are those whose charts imply low future normalized excess returns.
8. The paper then asks whether the resulting long-short return can be explained by risk, small-stock effects, unstable model training, momentum/reversal, image-based chart ML, technical indicators, or market frictions.

Interpretation: The paper should be read as a disciplined machine-learning test of weak-form efficiency. Its strongest feature is that the information set is deliberately narrow, while the forecasting function is flexible.

2 Data and measurement

2.1 Stock universe and periods

- The stock universe is U.S.-based common shares listed on the NYSE, AMEX, or NASDAQ.
- Each stock-month observation must have returns for months $t-12$ through $t-1$ and market capitalization at the end of month $t-1$.
- The **optimization period** is 1927:01–1963:06.
- The **test period** is 1963:07–2022:12.
- The average test-period month contains about 4,166 stocks.

Interpretation: The strict period split prevents the main asset-pricing results from being contaminated by choices made after looking at the test period.

2.2 Dependent variable: excess return and transformations

The raw dependent variable is the month-t excess return:

$$r_{i,t} = R_{i,t} - R_t^f, \tag{1}$$

where $R_{i,t}$ is the delisting-adjusted stock return and R_t^f is the risk-free rate.

The paper considers four dependent variables for model optimization:

- raw excess return r;
- standardized excess return r^{Std};
- normalized excess return r^{Norm};
- return percentile r^{Pctl}.

Interpretation: The chosen dependent variable is normalized excess return. Normalization makes the forecast focus on cross-sectional ranking rather than the magnitude of market-wide monthly moves.

2.3 Input variables: cumulative returns in a one-year chart

For each stock i and return month t, the input vector is

$$X_{i,t-1} = (\mathrm{CR}_{1,i,t}, \mathrm{CR}_{2,i,t}, \ldots, \mathrm{CR}_{12,i,t}). \tag{2}$$

For $k \in \{1, \ldots, 12\}$,

$$\mathrm{CR}_{k,i,t} = \prod_{j=12}^{12-k+1} (1 + R_{i,t-j}) - 1, \tag{3}$$

which is the cumulative return over the first k months in the one-year price plot ending at $t-1$.

Interpretation: CR_1 is the first monthly return shown in the chart, and CR_{12} is the cumulative return over the full twelve-month chart. The sequence encodes both shape and magnitude.

2.4 ML-based forecast

The machine-learning process learns a function

$$f : \mathbb{R}^{12} \to \mathbb{R} \tag{4}$$

that maps past cumulative returns into a forecast of future normalized excess return.

The forecast is

$$\mathrm{MLER}_{i,t} = f(\mathrm{CR}_{1,i,t}, \ldots, \mathrm{CR}_{12,i,t}). \tag{5}$$

Interpretation: MLER is not an observed stock characteristic. It is the output of the optimized forecasting function applied to a stock's chart.

2.5 Optimization design

The authors compare 96 ML implementation combinations:

$$4 \text{ architectures} \times 2 \text{ loss functions} \times 3 \text{ weighting schemes} \times 4 \text{ dependent variables} = 96. \tag{6}$$

The four architectures are FNN, CNN, LSTM, and CNNLSTM. The two loss functions are

$$\mathcal{L}_{MSE} = \sum_j w_j \epsilon_j^2, \qquad \mathcal{L}_{MAE} = \sum_j w_j |\epsilon_j|. \tag{7}$$

The three weighting schemes are equal weight by observation, equal weight by month and observation, and equal weight by month with within-month value weights.

Interpretation: The final specification is not chosen from test-period performance. It is chosen from rank correlations between forecasts and realized returns in the optimization period.

2.6 Expanding-window forecasts

For a fitting window $[t_1, t_2]$, define

$$\mathrm{MLER}_{i,t}^{t_1,t_2} \tag{8}$$

as the average of 30 forecasts produced by 30 separate neural-network fits using observations from t_1 to t_2.

The focal forecast is built from expanding windows:

$$\text{MLER}_{i,t} = \begin{cases} \text{MLER}_{i,t}^{192701,196306}, & 196307 \le t \le 197412, \\ \text{MLER}_{i,t}^{192701,197412}, & 197501 \le t \le 198412, \\ \text{MLER}_{i,t}^{192701,198412}, & 198501 \le t \le 199412, \\ \text{MLER}_{i,t}^{192701,199412}, & 199501 \le t \le 200412, \\ \text{MLER}_{i,t}^{192701,200412}, & 200501 \le t \le 201412, \\ \text{MLER}_{i,t}^{192701,201412}, & 201501 \le t \le 202212. \end{cases} \tag{9}$$

Interpretation: The forecasts are always out of sample relative to the period being predicted. Later forecasts use more historical data, mimicking a researcher or investor who updates the model over time.

3 Models and empirical specifications

3.1 Portfolio sorting test

Each month, stocks are sorted into ten portfolios using NYSE breakpoints of MLER. The return on portfolio d is

$$R_{d,t} = \sum_{i \in d_t} w_{i,t-1} R_{i,t}, \tag{10}$$

where $w_{i,t-1}$ is based on market capitalization at the end of month $t-1$.

The main long-short portfolio is

$$R_t^{10-1} = R_{10,t} - R_{1,t}. \tag{11}$$

Interpretation: The EMH predicts no systematic difference between high- and low-MLER stocks. The paper finds a large positive spread.

3.2 Risk-adjusted regressions

For each factor model, the long-short return is regressed on factors:

$$R_t^{10-1} = \alpha + \beta' F_t + \epsilon_t. \tag{12}$$

The factor sets include CAPM, Fama–French, Carhart, and Carhart augmented with a reversal factor.

Interpretation: If MLER is just loading on known risk factors, α should vanish. It remains large.

3.3 Subperiod and large-stock tests

The authors repeat the decile portfolio analysis by subperiod and by large-stock subsets. Large-stock tests vary:

- whether breakpoints are calculated using NYSE stocks or large stocks;
- whether the held stocks are the largest 500 firms or firms above NYSE size breakpoints.

Interpretation: Strong performance among large stocks makes microcap illiquidity or small-stock anomalies less plausible explanations.

3.4 Forecasting-function stability

For different historical fit windows, the authors compare:

- Spearman rank correlations between forecast signals;
- overlap in decile 1 and decile 10 holdings;
- correlations of long-short portfolio returns.

Interpretation: Stability matters because an unstable forecasting function would look more like overfitting than a persistent chart pattern.

3.5 Fama–MacBeth tests for nonlinearities and interactions

The paper examines whether MLER is explainable by linear functions of the inputs. A representative cross-sectional regression is

$$\text{MLER}_{i,t} = a_t + \sum_{k=1}^{12} b_{k,t}\text{CR}_{k,i,t} + u_{i,t}. \tag{13}$$

To capture nonlinearities, they use

$$\text{MLER}_{i,t} = a_t + \sum_{k=1}^{12} b_{k,t}\text{CR}_{k,i,t} + \sum_{k=1}^{12} c_{k,t}\text{MLER}_{\text{CR}_k,i,t} + u_{i,t}. \tag{14}$$

To capture interactions, they use versions in which all but one input is fixed at percentile values.

Interpretation: The core finding is that simple linear combinations of the 12 cumulative returns explain only part of MLER, and the residual nonlinear/interaction component matters for return prediction.

3.6 Momentum and reversal controls

Momentum and reversal are defined as

$$\text{Mom}_{i,t} = \prod_{j=2}^{12}(1 + R_{i,t-j}) - 1, \qquad \text{Rev}_{i,t} = R_{i,t-1}. \tag{15}$$

The paper also constructs an ML forecast based only on momentum and reversal:

$$\text{MLER}^{\text{Mom,Rev}}_{i,t} = g(\text{Mom}_{i,t}, \text{Rev}_{i,t}). \tag{16}$$

Interpretation: Comparing MLER to Mom, Rev, and $\text{MLER}^{\text{Mom,Rev}}$ separates chart patterns from the two most familiar return-based anomalies.

3.7 Technical-signal and image-based ML controls

The paper tests whether MLER is subsumed by:

- image-based ML forecasts from Jiang et al. (2022);
- 14 technical signals studied by Neely et al. (2014);
- 14 trading-friction variables from Freyberger et al. (2020).

Interpretation: The signal remains strong after these controls, suggesting that the machine chartist is detecting patterns different from known technical rules and alternative ML chart images.

3.8 Optimization-performance validation

For each candidate ML model m, the authors compare:

$$Perf_m^{opt} \quad \text{and} \quad Perf_m^{test}, \tag{17}$$

where optimization-period performance is based on Table 1 and test-period performance is the out-of-sample long-short return in Table 17.

Interpretation: A positive relation between optimization and test performance indicates that the optimization process selects genuinely useful models rather than overfit noise.

4 Identification and empirical design

4.1 What identifies predictive power

- The model-selection step is performed using only 1927:01–1963:06.
- The main return tests use 1963:07–2022:12.
- Forecasts in later periods are generated using expanding windows that end before the forecasted returns.
- Portfolio tests sort stocks only on information available at the end of $t-1$.

Interpretation: The identifying logic is out-of-sample prediction. The paper's strongest defense against data mining is the strict separation of optimization and test periods.

4.2 Why the result challenges weak-form efficiency

- Inputs are only historical stock returns.
- Historical stock returns are visible in price charts.
- The forecasts predict future cross-sectional returns strongly.
- The signal is not subsumed by standard momentum, reversal, or existing technical-analysis signals.

Interpretation: The evidence is directly targeted at weak-form EMH because the predictive information comes from past price behavior.

4.3 Why the design is not just standard anomaly mining

- The predictor is not hand selected from the test-period data.
- The authors compare many ML implementations using only an early optimization period.
- The final t-statistics exceed data-mining thresholds suggested by Harvey et al.
- The forecast works in large stocks and recent subperiods.

Interpretation: The concern is not eliminated completely, but the design is much more disciplined than searching many hand-crafted return patterns in the full sample.

4.4 What remains imperfect

- The model is a black-box predictor, so economic interpretation of every pattern is difficult.
- Neural-network choices still require judgment even when chosen in an optimization period.
- Transaction costs, capacity, and implementation frictions are not the paper's central focus.
- The evidence shows predictability, but it does not fully explain why the predictability persists.

Interpretation: The paper's contribution is primarily empirical: it documents strong chart-based predictability learned by machines.

5 Tables and figures

5.1 Figure 1 and Table 1: chart information and model optimization

Read:

- Figure 1 illustrates why raw returns are a problematic dependent variable: market-wide episodes such as 1932–1933 and 1930–1931 create aggregate chart patterns that could dominate training.
- Table 1 compares 96 combinations of architecture, loss, weighting, and dependent variable during the optimization period.
- The best-performing configuration uses CNNLSTM, MSE, equal weight per month and per stock, and normalized excess return.

Economic message: The paper deliberately selects the model before the test period, and the selected model is built to forecast cross-sectional rankings rather than aggregate market episodes.

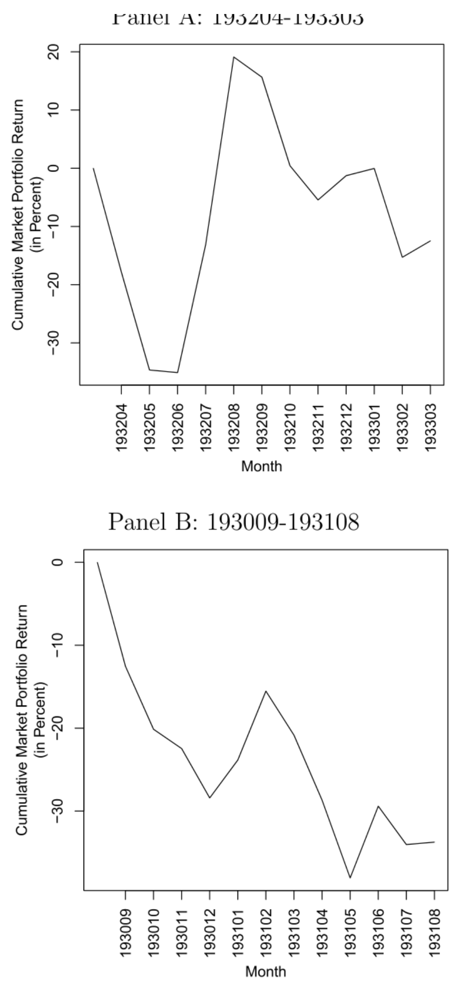


Figure 1: market portfolio price plots

Table 1

ML Model Optimization. This table presents the results of tests used to determine the optimal ML model. The tests are run using only data from the 192701-196306 optimization period. We examine all combinations of four ML architectures, two loss functions, three loss function weighting methodologies, and four dependent variables. The ML architectures are feed-forward neural network (FNN), convolutional neural network (CNN), long short-term memory (LSTM), and convolutional neural network with long short-term memory (CNNLSTM). The loss functions are the mean squared error (MSE) and the mean absolute error (MAE). The loss function weighting methodologies are to equal-weight each observation (EW), to equal-weight each month and within each month give equal weight to each observation (EWPM), and to equal-weight each month and within each month weight each observation according to its market capitalization (EWPMVW). The four dependent variables we examine are the excess stock return (r), the standardized excess stock return (r_{Std}), the normalized excess stock return (r_{Norm}), and the percentile of the stock return (r_{Pctl}). Using each of the 96 possible combinations of implementation choices, we train each model 30 times using data from the even months in even years and odd months in odd years, to generate 30 forecasting functions. We then apply each of the forecasting functions to each observation in odd months in even years and even months in odd years, and for each such observation, take the average of the 30 resulting values to be the forecast. The table shows the time-series averages of the monthly cross-sectional Spearman rank correlations between the forecasts and the actual excess return. The column labeled "Dependent Variable" indicates the dependent variable. The column labeled "Weighting Methodology" indicates the weighting methodology. The remaining column headers indicate the ML architecture and loss function. The Spearman rank correlations are shown in percent.

Dependent Variable	Weighting Methodology	FNN MAE	FNN MSE	CNN MAE	CNN MSE	LSTM MAE	LSTM MSE	CNNLSTM MAE	CNNLSTM MSE
r	EW	4.6	1.6	4.0	1.0	5.5	−1.8	5.3	0.8
	EWPM	4.6	1.4	3.8	1.2	6.7	0.8	5.0	2.6
	EWPMVW	0.9	−0.5	0.5	−2.3	2.5	0.9	1.0	2.0
r_{Std}	EW	3.4	2.3	9.7	7.3	9.8	8.5	10.4	9.8
	EWPM	5.0	1.7	8.9	7.4	9.8	7.6	10.5	9.9
	EWPMVW	4.7	2.0	4.7	4.6	4.7	5.1	6.0	4.7
r_{Norm}	EW	7.5	1.5	9.7	8.8	10.3	10.2	10.4	10.7
	EWPM	6.9	1.4	9.1	8.0	10.1	10.2	10.6	10.8
	EWPMVW	3.7	0.4	4.3	4.3	5.2	6.3	6.9	7.5
r_{Pctl}	EW	−2.3	−1.6	7.7	−2.7	9.8	9.4	10.2	10.1
	EWPM	0.7	−3.0	7.8	−3.5	9.6	9.3	10.2	10.2
	EWPMVW	−0.1	−3.1	1.2	−1.7	4.9	5.7	7.3	7.8

Table 1: ML model optimization

5.2 Table 2 and Table 3: forecast distribution and main portfolio return result

Read:

- Table 2 shows that MLER has an average cross-sectional standard deviation of about 0.11 and a near-zero median.

- Table 3 shows nearly monotonic returns across MLER deciles.
- The decile 10-minus-1 portfolio earns 1.08% per month with a t-statistic of 5.51 and Sharpe ratio of 0.78.

Economic message: The forecasts are not merely well calibrated in optimization. They produce large realized return spreads in the test period.

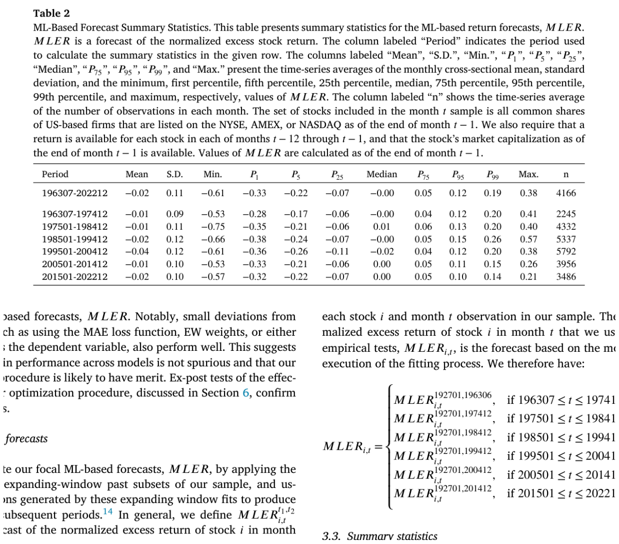


Table 2

ML-Based Forecast Summary Statistics. This table presents summary statistics for the ML-based return forecasts, $MLER$. $MLER$ is a forecast of the normalized excess stock return. The column labeled "Period" indicates the period used to calculate the summary statistics in the given row. The columns labeled "Mean", "S.D.", "Min.", "P_1", "P_5", "P_{25}", "Median", "P_{75}", "P_{95}", "P_{99}", and "Max." present the time-series averages of the monthly cross-sectional mean, standard deviation, and the minimum, first percentile, fifth percentile, 25th percentile, median, 75th percentile, 95th percentile, 99th percentile, and maximum, respectively, values of $MLER$. The column labeled "n" shows the time-series average of the number of observations in each month. The set of stocks included in the month t sample is all common shares of US-based firms that are listed on the NYSE, AMEX, or NASDAQ as of the end of month $t-1$. We also require that a return is available for each stock in each of months $t-12$ through $t-1$, and that the stock's market capitalization as of the end of month $t-1$ is available. Values of $MLER$ are calculated as of the end of month $t-1$.

Period	Mean	S.D.	Min.	P_1	P_5	P_{25}	Median	P_{75}	P_{95}	P_{99}	Max.	n
196307-202212	−0.02	0.11	−0.61	−0.33	−0.22	−0.07	−0.00	0.05	0.12	0.19	0.38	4166
196307-197412	−0.01	0.09	−0.53	−0.28	−0.17	−0.06	−0.00	0.04	0.12	0.20	0.41	2245
197501-198412	−0.01	0.11	−0.75	−0.35	−0.21	−0.06	0.01	0.06	0.13	0.20	0.40	4332
198501-199412	−0.02	0.12	−0.66	−0.38	−0.24	−0.07	−0.00	0.05	0.15	0.26	0.57	5337
199501-200412	−0.04	0.12	−0.61	−0.36	−0.26	−0.11	−0.02	0.04	0.12	0.20	0.38	5792
200501-201412	−0.01	0.10	−0.53	−0.33	−0.21	−0.06	0.00	0.05	0.11	0.15	0.26	3956
201501-202212	−0.02	0.10	−0.57	−0.32	−0.22	−0.07	0.00	0.05	0.10	0.14	0.21	3486

ɔased forecasts, $MLER$. Notably, small deviations from ch as using the MAE loss function, EW weights, or either ; the dependent variable, also perform well. This suggests in performance across models is not spurious and that our ɪrocedure is likely to have merit. Ex-post tests of the effec- : optimization procedure, discussed in Section 6, confirm s.

forecasts

te our focal ML-based forecasts, $MLER$, by applying the expanding-window past subsets of our sample, and us- ɔns generated by these expanding window fits to produce ;ubsequent periods.[14] In general, we define $MLER_{i,t}^{t_1,t_2}$ cast of the normalized excess return of stock i in month

each stock i and month t observation in our sample. The malized excess return of stock i in month t that we us empirical tests, $MLER_{i,t}$, is the forecast based on the mo execution of the fitting process. We therefore have:

$$MLER_{i,t} = \begin{cases} MLER_{i,t}^{192701,196306}, & \text{if } 196307 \le t \le 19741 \\ MLER_{i,t}^{192701,197412}, & \text{if } 197501 \le t \le 19841 \\ MLER_{i,t}^{192701,198412}, & \text{if } 198501 \le t \le 19941 \\ MLER_{i,t}^{192701,199412}, & \text{if } 199501 \le t \le 20041 \\ MLER_{i,t}^{192701,200412}, & \text{if } 200501 \le t \le 20141 \\ MLER_{i,t}^{192701,201412}, & \text{if } 201501 \le t \le 20221 \end{cases}$$

3.3. Summary statistics

Table 2: ML-based forecast summary statistics

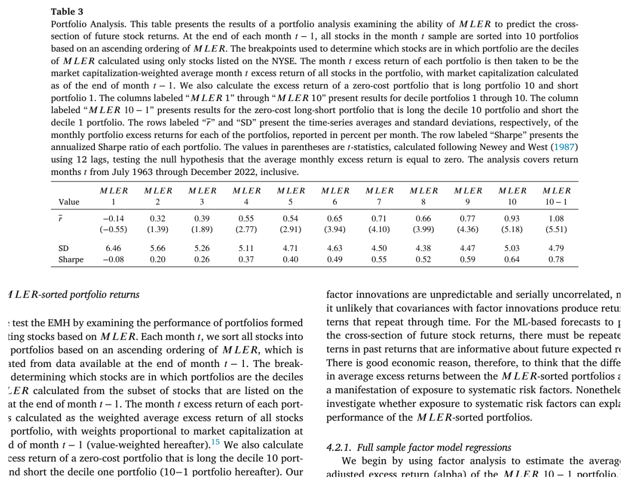


Table 3

Portfolio Analysis. This table presents the results of a portfolio analysis examining the ability of $MLER$ to predict the cross-section of future stock returns. At the end of each month $t-1$, all stocks in the month t sample are sorted into 10 portfolios based on an ascending ordering of $MLER$. The breakpoints used to determine which stocks are in which portfolio are the deciles of $MLER$ calculated using only stocks listed on the NYSE. The month t excess return of each portfolio is then taken to be the market capitalization-weighted average month t excess return of all stocks in the portfolio, with market capitalization calculated as of the end of month $t-1$. We also calculate the excess return of a zero-cost portfolio that is long portfolio 10 and short portfolio 1. The columns labeled "$MLER$ 1" through "$MLER$ 10" present results for decile portfolios 1 through 10. The column labeled "$MLER$ 10 − 1" presents results for the zero-cost long-short portfolio that is long the decile 10 portfolio and short the decile 1 portfolio. The rows labeled "$\bar{r}$" and "SD" present the time-series averages and standard deviations, respectively, of the monthly portfolio excess returns for each of the portfolios, reported in percent per month. The row labeled "Sharpe" presents the annualized Sharpe ratio of each portfolio. The values in parentheses are t-statistics, calculated following Newey and West (1987) using 12 lags, testing the null hypothesis that the average monthly excess return is equal to zero. The analysis covers return months t from July 1963 through December 2022, inclusive.

Value	$MLER$ 1	$MLER$ 2	$MLER$ 3	$MLER$ 4	$MLER$ 5	$MLER$ 6	$MLER$ 7	$MLER$ 8	$MLER$ 9	$MLER$ 10	$MLER$ 10 − 1
$\bar{r}$	−0.14	0.32	0.39	0.55	0.54	0.65	0.71	0.66	0.77	0.93	1.08
	(−0.55)	(1.39)	(1.89)	(2.77)	(2.91)	(3.94)	(4.10)	(3.99)	(4.36)	(5.18)	(5.51)
SD	6.46	5.66	5.26	5.11	4.71	4.63	4.50	4.38	4.47	5.03	4.79
Sharpe	−0.08	0.20	0.26	0.37	0.40	0.49	0.55	0.52	0.59	0.64	0.78

ɪLER-sorted portfolio returns

: test the EMH by examining the performance of portfolios formed ting stocks based on $MLER$. Each month t, we sort all stocks into portfolios based on an ascending ordering of $MLER$, which is ated from data available at the end of month $t-1$. The break- determining which stocks are in which portfolios are the deciles ĿER calculated from the subset of stocks that are listed on the at the end of month $t-1$. The month t excess return of each port- s calculated as the weighted average excess return of all stocks portfolio, with weights proportional to market capitalization at d of month $t-1$ (value-weighted hereafter).[15] We also calculate cess return of a zero-cost portfolio that is long the decile 10 port- nd short the decile one portfolio (10−1 portfolio hereafter). Our

factor innovations are unpredictable and serially uncorrelated, n it unlikely that covariances with factor innovations produce retu terns that repeat through time. For the ML-based forecasts to p the cross-section of future stock returns, there must be repeate terns in past returns that are informative about future expected re There is good economic reason, therefore, to think that the diffe in average excess returns between the $MLER$-sorted portfolios a a manifestation of exposure to systematic risk factors. Nonethele investigate whether exposure to systematic risk factors can expla performance of the $MLER$-sorted portfolios.

4.2.1. Full sample factor model regressions

We begin by using factor analysis to estimate the averag adjusted excess return (alpha) of the $MLER$ 10 − 1 portfolio.

Table 3: main portfolio analysis

5.3 Table 4, Table 5, and Figure 2: risk adjustment and time variation

Read:

- Table 4 shows that factor-model alphas of the long-short portfolio remain large and significant under CAPM, FF, FFC, and FFC plus reversal.
- The risk metrics in Table 4 do not point to a standard risk compensation story.
- Table 5 shows that the long-short portfolio works in most subperiods, though performance is weaker in 2005–2014.
- Figure 2 shows the cumulative log excess return of the MLER long-short portfolio through the test period.

Economic message: The return spread is not simply compensation for known risk-factor exposure and is not confined to one short historical episode.

LER 5	MLER 6	MLER 7	MLER 8	MLER 9	MLER 10	MLER 10 − 1
).20	−0.03	0.15	0.07	−0.03	0.25	1.15
).50)	(−0.09)	(0.33)	(0.17)	(−0.07)	(0.62)	(3.10)
).15	−0.03	0.12	0.06	−0.02	0.17	1.16
.65	0.66	0.85	0.81	1.03	1.47	1.32
.46)	(1.60)	(1.90)	(2.00)	(2.28)	(3.13)	(4.41)
.50	0.50	0.63	0.62	0.73	0.93	1.15
.85	0.78	0.79	0.84	0.98	1.00	1.17
.49)	(2.49)	(2.67)	(2.68)	(3.19)	(2.77)	(4.07)
.65	0.59	0.59	0.63	0.71	0.67	1.16
.55	0.93	0.72	0.94	1.10	1.57	1.71
.14)	(2.28)	(1.52)	(2.60)	(2.51)	(4.06)	(3.21)
.40	0.67	0.58	0.83	0.96	1.16	0.90
.59	0.97	0.88	0.64	0.56	0.40	−0.08
.19)	(2.35)	(2.13)	(1.28)	(1.18)	(0.71)	(−0.15)
.45	0.76	0.72	0.52	0.46	0.27	−0.05
.00	0.70	1.03	0.77	1.17	1.01	1.20
.83)	(1.40)	(2.16)	(2.04)	(2.99)	(3.16)	(2.13)
.61	0.46	0.73	0.58	0.85	0.78	0.75

Table 4

Risk-Adjusted Returns. This table presents the results of analyses examining whether the average returns of the portfolios formed by sorting on ML-based forecasts are compensation for risk. Panel A presents the results of factor model regressions of the excess returns of the $MLER$ 10 − 1 portfolio on the excess returns of the factors in different factor models. The $MLER$ 10 − 1 portfolio is the same as that whose average excess returns are shown in Table 3. The rows labeled "α", "$SD(\epsilon)$", and "Adj. R^2" present the intercept, residual standard deviation, and adjusted R^2, respectively, of the regression. The row labeled "Sharpe($\alpha + \epsilon$)" presents the annualized Sharpe ratio of the portfolio that is long the $MLER$ 10 − 1 portfolio and short the factor portfolios in amounts dictated by the slope coefficients estimated by the regression, calculated as the alpha divided by the residual standard deviation, times $\sqrt{12}$. Panel B presents results from 6-month non-overlapping subperiod factor model regressions estimated using daily excess return data. The row labeled "α" presents the average monthly alphas (intercept coefficients multiplied by 21) from these regressions. The row labeled "$SD(\epsilon)$" presents the standard deviation of the residuals from these regressions times $\sqrt{21}$. The standard deviation of the residuals from these regressions is calculated by using the residuals from the individual short subperiod regressions to generate a full time-series of daily residuals for the entire 196301-202212 period and calculating the standard deviation of this full time-series of residuals. The row labeled "Sharpe($\alpha + \epsilon$)" presents the annualized Sharpe ratio of the portfolio, calculated by dividing the average monthly alpha by the monthly standard deviation of the residuals and multiplying by $\sqrt{12}$. The row labeled "Adj. R^2" presents the average adjusted R^2 value from the short subperiod regressions. Except for the column labeled "Value", the column headers indicate the factor model used to generate the results in the given column. All excess returns, standard deviations, and alphas are reported in percent per month. The values in parentheses are t-statistics testing the null hypothesis that the average monthly alpha is equal to zero. The t-statistics in Panel A are calculated following Newey and West (1987) using 12 lags. The analyses cover return months t from July 1963 through December 2022, inclusive.

Panel A: $MLER$ 10 − 1 Portfolio Full Sample Factor Regressions

Value	CAPM	FF	FFC	FFC+REV	FF5	Q
α	1.20	1.24	0.79	0.45	1.02	0.71
	(6.43)	(7.28)	(5.23)	(3.58)	(6.67)	(3.78)
$SD(\epsilon)$	4.68	4.51	4.00	3.41	4.38	4.30
Sharpe($\alpha + \epsilon$)	0.89	0.95	0.68	0.46	0.81	0.57
Adj. R^2	4.18%	10.89%	29.90%	49.05%	15.90%	22.15%

Panel B: $MLER$ 10 − 1 Portfolio Average Short Subperiod Factor Regressions

Value	CAPM	FF	FFC	FFC+REV	FF5	Q
α	1.30	1.39	1.48	0.56	1.30	1.20
	(8.46)	(9.56)	(11.57)	(3.77)	(9.51)	(9.32)
$SD(\epsilon)$	4.27	3.75	3.45	3.24	3.57	3.60
Sharpe($\alpha + \epsilon$)	1.06	1.28	1.49	0.59	1.26	1.15
Adj. R^2	10.42%	25.22%	32.02%	37.36%	29.28%	28.74%

Table 4: risk-adjusted returns

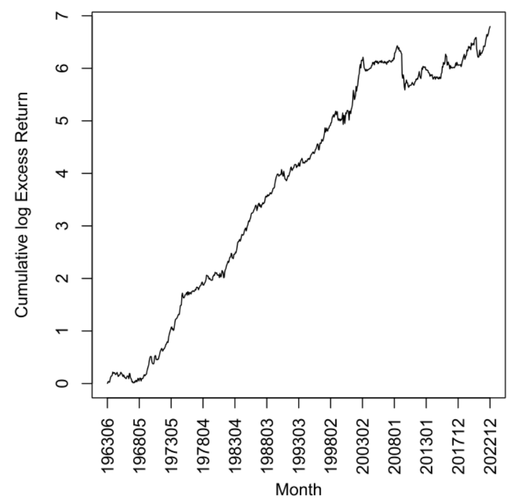


Figure 2: cumulative performance

Table 5

Portfolio Subperiod Analysis. This table describes the performance of the $MLER$ decile portfolios during different subperiods. The portfolios are the exact same portfolios whose performances are examined in Table 3. The column labeled "Subperiod" indicates the subperiod covered by each analysis. The columns labeled "$MLER$ 1" through "$MLER$ 10" and "$MLER$ 10 − 1" present results for the portfolio indicated in the column header. The rows labeled "$\bar{r}$" and "Sharpe" present the time-series averages of the monthly portfolio excess returns, reported in percent per month, and the annualized Sharpe ratios, respectively, for each of the portfolios in the given period. The values in parentheses are t-statistics, calculated following Newey and West (1987) using 12 lags, testing the null hypothesis that the average monthly excess return is equal to zero.

Subperiod	Value	MLER 1	MLER 2	MLER 3	MLER 4	MLER 5	MLER 6	MLER 7	MLER 8	MLER 9	MLER 10	MLER 10 − 1
196307-197412	$\bar{r}$	−0.90	−0.47	−0.26	−0.14	−0.20	−0.03	0.15	0.07	−0.03	0.25	1.15
		(−1.39)	(−0.84)	(−0.49)	(−0.32)	(−0.50)	(−0.09)	(0.33)	(0.17)	(−0.07)	(0.62)	(3.10)
	Sharpe	−0.56	−0.32	−0.19	−0.10	−0.15	−0.03	0.12	0.06	−0.02	0.17	1.16
197501-198412	$\bar{r}$	0.16	0.27	0.42	0.65	0.65	0.66	0.85	0.81	1.03	1.47	1.32
		(0.31)	(0.58)	(0.89)	(1.38)	(1.46)	(1.60)	(1.90)	(2.00)	(2.28)	(3.13)	(4.41)
	Sharpe	0.09	0.17	0.30	0.47	0.50	0.50	0.63	0.62	0.73	0.93	1.15
198501-199412	$\bar{r}$	−0.17	0.49	0.44	0.63	0.85	0.78	0.79	0.84	0.98	1.00	1.17
		(−0.57)	(1.51)	(1.54)	(2.08)	(2.49)	(2.49)	(2.67)	(2.68)	(3.19)	(2.77)	(4.07)
	Sharpe	−0.13	0.37	0.34	0.51	0.65	0.59	0.59	0.63	0.71	0.67	1.16
1995012-20041	$\bar{r}$	−0.14	0.66	0.50	0.68	0.55	0.93	0.72	0.94	1.10	1.57	1.71
		(−0.18)	(1.15)	(0.96)	(1.17)	(1.14)	(2.28)	(1.52)	(2.60)	(2.51)	(4.06)	(3.21)
	Sharpe	−0.06	0.35	0.30	0.41	0.40	0.67	0.58	0.83	0.96	1.16	0.90
200501-201412	$\bar{r}$	0.48	0.49	0.91	0.62	0.59	0.97	0.88	0.64	0.56	0.40	−0.08
		(0.67)	(0.67)	(1.51)	(1.09)	(1.19)	(2.35)	(2.13)	(1.28)	(1.18)	(0.71)	(−0.15)
	Sharpe	0.24	0.27	0.57	0.42	0.45	0.76	0.72	0.52	0.46	0.27	−0.05

5.4 Tables 6–9: large stocks and stability of the forecasting relation

Read:

- Table 6 shows that the forecast remains useful among large stocks, including samples close to the largest 500 firms.
- Table 7 shows high correlations and overlap among forecasts generated from different historical windows.
- Table 8 shows that forecasts trained in specific subperiods predict returns in other periods.
- Table 9 shows that expanding-window and rolling-window forecasts both produce strong portfolio performance.

Economic message: The effect is not a microcap artifact, and the forecasting function is stable enough to look like a persistent empirical relation rather than a one-time overfit.

Table 6

Predictive Power Among Large Stocks. This table presents the results of portfolio analyses examining the ability of the ML-based forecasts to predict the cross-section of future stock returns among large stocks. The construction of the portfolios is exactly the same as was used to construct the portfolios whose performances are examined in Table 3, except that we vary the set of stocks used in the calculation of the portfolio breakpoints and the set of stocks that are sorted into the portfolios. The column labeled "Breakpoints" indicates the set of stocks used to calculate the breakpoints. The column labeled "Holdings" indicates the set of stocks sorted into the portfolios. The $Size > P_{20}^{\mathrm{NYSE}}$ (NYSE/$Size > P_{20}^{\mathrm{NYSE}}$) and $Size > P_{50}^{\mathrm{NYSE}}$ (NYSE/$Size > P_{50}^{\mathrm{NYSE}}$) subsets include all (NYSE-listed) stocks with market capitalizations greater than the 20th and 50th percentile values, respectively, among NYSE-listed stocks as of the end of month $t-1$. The Top 500 subset includes the top 500 stocks by market capitalization at the end of month $t-1$. The columns labeled "$MLER$ 1" through "$MLER$ 10" and "$MLER$ 10 − 1" present results for the portfolio indicated in the column header. The table shows the average excess return of each portfolio. All excess returns are reported in percent per month. The values in parentheses are t-statistics, calculated following Newey and West (1987) using 12 lags, testing the null hypothesis that the average monthly excess return is equal to zero. The analyses cover return months t from July 1963 through December 2022, inclusive.

Breakpoints	Holdings	$MLER$ 1	$MLER$ 2	$MLER$ 3	$MLER$ 4	$MLER$ 5	$MLER$ 6	$MLER$ 7	$MLER$ 8	$MLER$ 9	$MLER$ 10	$MLER$ 10 − 1
$Size > P_{20}^{\mathrm{NYSE}}$	$Size > P_{20}^{\mathrm{NYSE}}$	−0.13	0.34	0.39	0.57	0.53	0.60	0.75	0.62	0.75	0.92	1.06
		(−0.52)	(1.43)	(1.92)	(2.90)	(2.86)	(3.53)	(4.37)	(3.78)	(4.26)	(5.14)	(5.68)
NYSE/$Size > P_{20}^{\mathrm{NYSE}}$	$Size > P_{20}^{\mathrm{NYSE}}$	0.00	0.35	0.44	0.61	0.51	0.63	0.74	0.64	0.74	0.91	0.91
		(0.01)	(1.53)	(2.31)	(3.14)	(2.85)	(3.75)	(4.50)	(3.77)	(4.31)	(5.07)	(5.10)
$Size > P_{50}^{\mathrm{NYSE}}$	$Size > P_{50}^{\mathrm{NYSE}}$	0.04	0.33	0.48	0.52	0.52	0.61	0.64	0.62	0.77	0.85	0.81
		(0.17)	(1.52)	(2.34)	(2.78)	(2.92)	(3.68)	(3.72)	(3.70)	(4.52)	(4.60)	(4.63)
NYSE/$Size > P_{50}^{\mathrm{NYSE}}$	$Size > P_{50}^{\mathrm{NYSE}}$	0.10	0.37	0.46	0.61	0.55	0.62	0.66	0.66	0.75	0.87	0.77
		(0.42)	(1.80)	(2.27)	(3.38)	(3.21)	(3.63)	(3.97)	(3.91)	(4.37)	(4.82)	(4.66)
Top 500	Top 500	0.10	0.43	0.37	0.58	0.48	0.64	0.58	0.66	0.73	0.82	0.72
		(0.41)	(1.93)	(1.88)	(3.10)	(2.95)	(3.82)	(3.26)	(3.91)	(4.14)	(4.56)	(4.37)

Predictive power for large stocks

Recent work shows that many return anomalies are concentrated
ɔng small stocks (Fama and French (2008, 2018), Hou et al. (2020))
that trading costs make such anomalies unprofitable to trade (Novy-
rx and Velikov (2016), Detzel et al. (2023)). The portfolios examined
his point are value-weighted and constructed using breakpoints cal-
ıted from only NYSE-listed stocks. Hou et al. (2020) show that this
roach minimizes the influence of small stocks on the results of the
lyses. Nonetheless, to ensure that our results persist when small
:ks are excluded, we repeat the portfolio analysis using subsets of
sample that exclude small stocks from both the set of stocks held
he portfolios and the set of stocks used to calculate the breakpoints.
define the $Size > P_{20}^{\mathrm{NYSE}}$ (NYSE/$Size > P_{20}^{\mathrm{NYSE}}$) and $Size > P_{50}^{\mathrm{NYSE}}$
'SE/$Size > P_{50}^{\mathrm{NYSE}}$) subsets to be the sets of (NYSE-listed) stocks with
:ket capitalizations greater than the 20th and 50th percentile val-
, respectively, among NYSE-listed stocks, and the Top 500 subset
nclude only the top 500 stocks by market capitalization. Each of
market capitalization-based criteria are evaluated monthly using all
:ks in the sample for the given month. We continue to use value-

the predictive power of the ML-based forecasts is strong among lar
stocks.

5. Characterization of ML-based forecasts

Having demonstrated that the ML-based forecasts have strong abili
to predict the cross-section of future stock returns, we proceed now
further characterize this predictive power.

5.1. Stability of forecasting relation

Our first tests characterizing the ML-based forecasts examine t
stability of the relation between past price patterns and future sto
returns (forecasting relation hereafter). Specifically, we investiga
whether price patterns associated with high (low) future stock retur
in the early part of our test period continue to be associated with hi
(low) stock returns for the duration of our test period. If the foreca
ing relation is stable, these relations could be learned over a prolong
period of time, thereby further suggesting the viability of charting.
To assess the stability of the forecasting relation, we defi

Table 6: predictive power among large stocks

Table 7

Stability of Forecasting Relation. This table presents the results of analyses examining the stability of the forecasting relation. We define $MLER^{t_1,t_2}$ as the ML-based forecast generated by applying the ML process to data from months t between t_1 and t_2, inclusive. The columns under the "Forecast Correlations" heading present the time-series averages of monthly cross-sectional Spearman rank correlations between pairs of forecasts. The columns under the "10 − 1 Common Holdings" heading present the time-series averages of the monthly percentages, in decimal format (i.e. 0.01 = 1%) of holdings in the 10 − 1 portfolios that are common to both portfolios. The percentage of common holdings in any pair of 10 − 1 portfolios in any month is the number of stocks held in the same direction (both long or both short) in both portfolios divided by the average of the number of stocks in the portfolios. The columns under the "10 − 1 Return Correlations" heading present the time-series Pearson product-moment correlations between the excess returns of the 10 − 1 portfolios. With the exception of the sort variable, the 10 − 1 portfolios are constructed in exactly the same manner as those whose returns are examined in Table 3. All analyses use data from all months in the 196307-202212 test period except those months whose data are used by the ML process to generate the forecasting function upon which either forecast is based.

	Forecast Correlations					10 − 1 Common Holdings					10 − 1 Return Correlations				
	$MLER^{196307,197412}$	$MLER^{197501,198412}$	$MLER^{198501,199412}$	$MLER^{199501,200412}$	$MLER^{200501,201412}$	$MLER^{196307,197412}$	$MLER^{197501,198412}$	$MLER^{198501,199412}$	$MLER^{199501,200412}$	$MLER^{200501,201412}$	$MLER^{196307,197412}$	$MLER^{197501,198412}$	$MLER^{198501,199412}$	$MLER^{199501,200412}$	$MLER^{200501,201412}$
$MLER^{192701,196306}$	0.62	0.56	0.50	0.49	0.31	0.50	0.48	0.41	0.41	0.30	0.60	0.54	0.37	0.41	0.30
$MLER^{196307,197412}$		0.74	0.73	0.71	0.57		0.60	0.58	0.57	0.45		0.84	0.79	0.80	0.66
$MLER^{197501,198412}$			0.80	0.78	0.68			0.64	0.62	0.52			0.84	0.88	0.81
$MLER^{198501,199412}$				0.84	0.77				0.65	0.58				0.91	0.85
$MLER^{199501,200412}$					0.75					0.56					0.83

Table 7: stability of forecasting relation

Table 8

Subperiod-Based Forecasts and Returns. This table presents the results of portfolio analyses examining the ability of ML-based forecasts generated by applying the ML process to data from one subperiod to predict the cross-section of stock returns during other subperiods. The column labeled "Return Period" indicates the period for which the excess portfolio returns are examined. The remaining column names indicate the variable used to construct the portfolios. $MLER^{t_1,t_2}$ is the ML-based forecast generated by applying the ML process to data from months t between t_1 and t_2, inclusive. With the exception of the sort variable, the portfolio construction methodology is identical to that used to construct the portfolios examined in Table 3. The table presents the average monthly excess return of the 10 − 1 portfolio formed by sorting on the given variable during the given subperiod. All excess returns are reported in percent per month. The values in parentheses are t-statistics, calculated following Newey and West (1987) using 12 lags, testing the null hypothesis that the average monthly excess return is equal to zero.

Return Period	$MLER^{192701,196306}$	$MLER^{196307,197412}$	$MLER^{197501,198412}$	$MLER^{198501,199412}$	$MLER^{199501,200412}$	$MLER^{200501,201412}$

Table 8: subperiod-based forecasts

Table 9

Expanding-Window and Rolling-Window Forecast Portfolios. This table presents the results of portfolio analyses examining the ability of ML-based forecasts based on expanding window ($MLER$) and rolling-window ($MLER^{Roll}$) fit periods to predict the cross-section of future stock returns. With the exception of the sort variable for the portfolios formed by sorting on $MLER^{Roll}$, the portfolio construction methodology is identical to that used to construct the portfolios examined in Table 3. The column labeled "Sort Variable" indicates the variable used to sort stocks into portfolios. The columns labeled "1" through "10" present results for decile portfolios 1 through 10 formed by sorting on the given variable. The column labeled "10 − 1" presents results for the zero-cost long-short portfolio that is long the decile 10 portfolio and short the decile 1 portfolio. The rows with "$\bar{r}$" and "SD" in the column labeled "Value" present the time-series averages and standard deviations, respectively, of the monthly portfolio excess returns for each of the portfolios, reported in percent per month. The rows with "Sharpe" in the "Value" column present the annualized Sharpe ratio of the given portfolio. The values in parentheses are t-statistics, calculated following Newey and West (1987) using 12 lags, testing the null hypothesis that the average monthly excess return is equal to zero. The analyses cover return months t from July 1963 through December 2022, inclusive.

Sort Variable	Value	1	2	3	4	5	6	7	8	9	10	10 − 1
$MLER$	$\bar{r}$	−0.14	0.32	0.39	0.55	0.54	0.65	0.71	0.66	0.77	0.93	1.08
		(−0.55)	(1.39)	(1.89)	(2.77)	(2.91)	(3.94)	(4.10)	(3.99)	(4.36)	(5.18)	(5.51)
	SD	6.46	5.66	5.26	5.11	4.71	4.63	4.50	4.38	4.47	5.03	4.79
	Sharpe	−0.08	0.20	0.26	0.37	0.40	0.49	0.55	0.52	0.59	0.64	0.78
$MLER^{Roll}$	$\bar{r}$	−0.13	0.30	0.43	0.56	0.60	0.63	0.72	0.65	0.76	0.84	0.96
		(−0.43)	(1.21)	(1.83)	(2.68)	(3.32)	(3.46)	(4.43)	(4.01)	(4.60)	(4.96)	(4.24)
	SD	7.17	6.40	5.76	5.26	4.89	4.66	4.50	4.45	4.28	4.61	5.37
	Sharpe	−0.06	0.16	0.26	0.37	0.43	0.47	0.55	0.50	0.61	0.63	0.62

Table 9: rolling and expanding forecasts

5.5 Figures 3–4 and Table 10: nonlinearities and interactions

Read:

- Figure 3 shows univariate partial-dependence plots of MLER with respect to each cumulative-return input.

- Figure 4 shows bivariate partial-dependence plots, illustrating interactions such as between CR_4 and CR_{12}.
- Table 10 shows that linear functions of the 12 cumulative-return variables explain only part of MLER.
- Nonlinear and interaction components remain important for predicting future returns.

Economic message: The forecasting function is not just a smoothed version of momentum. Machine learning matters because the relation is nonlinear and interactive.

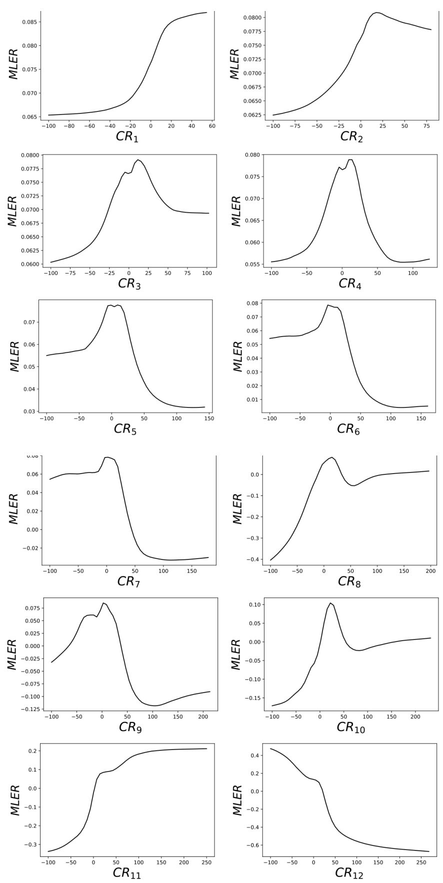


Figure 3: univariate partial-dependence plots

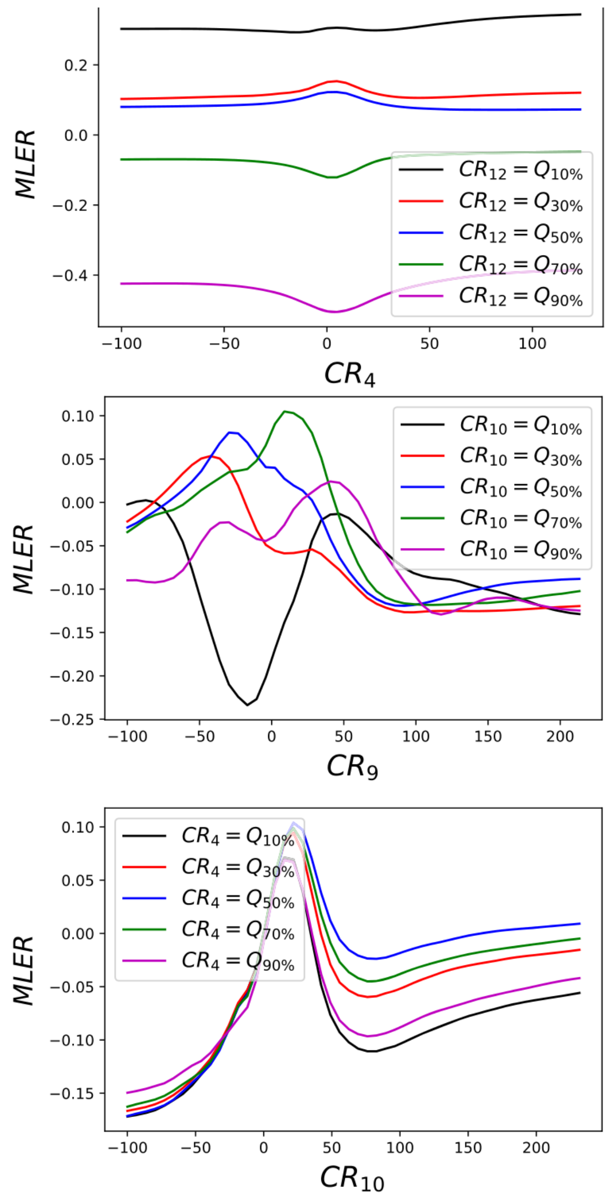


Figure 4: bivariate partial-dependence plots

Table 10

Nonlinearities and Interactions in the Forecasting Function. This table presents the results of Fama and MacBeth (1973) regression analyses examining nonlinearity and interactions of the ML-based forecasts $MLER$. Each month t we run a cross-sectional regression of the dependent variable on one or more independent variables. The results in Panel A are for regressions with $MLER$ as the dependent variable and $CR_1, \dots, CR_{12}$ as independent variables. The results in Panel B are for regressions with the future excess stock return as the dependent variable and $MLER$ and $CR_1, \dots, CR_{12}$ as independent variables. The results in Panel C are for regressions with $MLER$ as the dependent variable and $CR_1, \dots, CR_{12}$, and $MLER_{CR_1}, \dots, MLER_{CR_{12}}$ as independent variables. The results in Panel D are for regressions with the future excess stock return as the dependent variable and $MLER$, $CR_1, \dots, CR_{12}$, and $MLER_{CR_1}, \dots, MLER_{CR_{12}}$ as independent variables. $MLER_{CR_k}$ is the value of the ML-based forecast calculated using CR_k and the within-month means of CR_j for $j \in \{1, \dots, 12\}$ and $j \neq k$, as the values of the input variables, and therefore captures the component of the ML-based forecast that is nonlinear in CR_k. In Panels B and D, the sections with "EW" and "VW" in the "Weight" column present results for equal-weighted and market capitalization-weighted regressions, respectively. The table presents the time-series averages of the monthly cross-sectional coefficients, along with t-statistics, calculated following Newey and West (1987) using 12 lags, testing the null hypothesis that the average coefficient is equal to zero (in parentheses). The column labeled "Adj. R^2" in Panels A and C presents the time-series average of adjusted R^2 values from the monthly cross-sectional regressions. In Panels A and C, we report the estimated slope coefficients times 100. The analyses cover sample months t from July 1963 through December 2022, inclusive.

Panel A: FM Regressions of $MLER$ on $CR_1, \dots, CR_{12}$

CR_1	CR_2	CR_3	CR_4	CR_5	CR_6	
0.029	0.022	0.006	0.009	0.004	0.002	
(11.81)	(13.19)	(2.82)	(6.39)	(2.59)	(1.12)	
CR_7	CR_8	CR_9	CR_{10}	CR_{11}	CR_{12}	Adj. R^2
0.012	0.033	−0.065	0.131	0.251	−0.276	37.55%
(3.31)	(9.36)	(−11.18)	(16.94)	(17.00)	(−15.52)	

Panel B: FM Regressions of Future Excess Returns on $MLER$ and $CR_1, \dots, CR_{12}$

Weight	$MLER$	CR_1	CR_2	CR_3	CR_4	CR_5	CR_6
EW	5.95	0.01	0.00	−0.00	0.00	−0.00	0.00
	(9.06)	(2.21)	(0.76)	(−1.89)	(1.69)	(−1.62)	(0.13)
		CR_7	CR_8	CR_9	CR_{10}	CR_{11}	CR_{12}
		−0.00	−0.00	0.00	−0.00	0.02	−0.01
		(−0.90)	(−0.75)	(0.50)	(−0.67)	(4.31)	(−2.52)
	$MLER$	CR_1	CR_2	CR_3	CR_4	CR_5	CR_6
VW	2.81	0.00	0.00	−0.01	0.02	−0.00	−0.01
	(5.47)	(0.96)	(1.08)	(−3.66)	(4.23)	(−0.13)	(−2.97)
		CR_7	CR_8	CR_9	CR_{10}	CR_{11}	CR_{12}
		0.01	0.00	−0.00	0.00	0.02	−0.01
		(2.43)	(0.04)	(−1.19)	(0.54)	(3.84)	(−2.43)

Panel C: FM Regressions of $MLER$ on $CR_1, \dots, CR_{12}$, and $MLER_{CR_1}, \dots, MLER_{CR_{12}}$

CR_1	CR_2	CR_3	CR_4	CR_5	CR_6	
−0.009	0.009	0.001	0.008	0.003	−0.000	
(−2.61)	(5.04)	(0.30)	(6.56)	(2.09)	(−0.15)	
CR_7	CR_8	CR_9	CR_{10}	CR_{11}	CR_{12}	
0.011	0.029	−0.080	0.131	0.197	−0.235	
(3.29)	(8.14)	(−13.47)	(16.48)	(16.04)	(−19.62)	
$MLER_{CR_1}$	$MLER_{CR_2}$	$MLER_{CR_3}$	$MLER_{CR_4}$	$MLER_{CR_5}$	$MLER_{CR_6}$	
282.503	89.942	51.901	31.352	23.228	7.921	
(11.97)	(8.20)	(7.20)	(6.76)	(8.76)	(4.05)	
$MLER_{CR_7}$	$MLER_{CR_8}$	$MLER_{CR_9}$	$MLER_{CR_{10}}$	$MLER_{CR_{11}}$	$MLER_{CR_{12}}$	Adj. R^2
18.418	20.680	−18.183	1.162	30.575	15.374	55.36%
(5.39)	(14.28)	(−11.48)	(0.86)	(22.06)	(8.78)	

(*continued on next page*)

:t bivariate portfolios, each month t we sort stocks into
sed on an ascending ordering of the control variable.
ll stocks within each control variable-based group into
based on an ascending ordering of $MLER$. We take the
return for each of the resulting 25 portfolios to be the
l average month t excess return of all stocks in the port-
a single portfolio for each $MLER$ quintile, we define
cess return of the bivariate $MLER$ quintile k portfolio
l-weighted average of the month t excess returns of the

five $MLER$ quintile k portfolios (one for each control
tile). Finally, we define the month t bivariate $MLER$ 5
excess return to be the month t excess return of the biva
quintile five portfolio minus that of the bivariate MLEI
portfolio.

We construct the trivariate portfolios in a similar m
before sorting on $MLER$ we sort on both Mom and Rev
each month t we sort the stocks in our sample into 25
on the intersections of five Mom groups and five Rev

Table 10: nonlinearities and interactions, part 1

Table 10 (*continued*)

Panel D: FM Regressions of Future Excess Returns on $MLER$, $CR_1, \dots, CR_{12}$ and $MLER_{CR_1}, \dots, MLER_{CR_{12}}$

Weight	Slope Coefficients						
EW	$MLER$	CR_1	CR_2	CR_3	CR_4	CR_5	CR_6
	5.94	−0.00	0.00	−0.00	0.00	−0.00	0.00
	(10.29)	(−0.21)	(0.05)	(−2.09)	(1.12)	(−1.82)	(0.63)
		CR_7	CR_8	CR_9	CR_{10}	CR_{11}	CR_{12}
		−0.00	−0.00	0.00	−0.00	0.02	−0.01
		(−1.34)	(−0.50)	(0.37)	(−0.89)	(4.27)	(−2.63)
		$MLER_{CR_1}$	$MLER_{CR_2}$	$MLER_{CR_3}$	$MLER_{CR_4}$	$MLER_{CR_5}$	$MLER_{CR_6}$
		61.58	12.01	0.37	4.53	−8.79	4.35
		(1.68)	(1.00)	(0.04)	(0.96)	(−2.00)	(1.55)
		$MLER_{CR_7}$	$MLER_{CR_8}$	$MLER_{CR_9}$	$MLER_{CR_{10}}$	$MLER_{CR_{11}}$	$MLER_{CR_{12}}$
		−1.17	−0.86	−0.15	−0.16	0.52	−0.00
		(−0.56)	(−1.14)	(−0.22)	(−0.26)	(1.28)	(−0.00)
VW	$MLER$	CR_1	CR_2	CR_3	CR_4	CR_5	CR_6
	3.08	0.00	0.00	−0.01	0.01	−0.00	−0.01
	(6.47)	(0.69)	(0.66)	(−3.05)	(3.84)	(−0.37)	(−2.86)
		CR_7	CR_8	CR_9	CR_{10}	CR_{11}	CR_{12}
		0.01	−0.00	−0.00	0.00	0.02	−0.01
		(2.12)	(−0.24)	(−0.38)	(0.51)	(3.79)	(−1.73)
		$MLER_{CR_1}$	$MLER_{CR_2}$	$MLER_{CR_3}$	$MLER_{CR_4}$	$MLER_{CR_5}$	$MLER_{CR_6}$
		31.43	1.42	−4.79	−1.14	−2.45	0.82
		(0.69)	(0.07)	(−0.44)	(−0.17)	(−0.40)	(0.20)
		$MLER_{CR_7}$	$MLER_{CR_8}$	$MLER_{CR_9}$	$MLER_{CR_{10}}$	$MLER_{CR_{11}}$	$MLER_{CR_{12}}$
		−0.04	−0.07	0.75	0.16	−0.11	0.87
		(−0.02)	(−0.07)	(0.72)	(0.18)	(−0.20)	(1.61)

Table 10: continued

5.6 Tables 11–12: momentum and reversal

Read:

- Table 11 shows that MLER is related to momentum and reversal, but these variables do not fully explain it.
- Table 12 shows that the predictive power of MLER survives bivariate and trivariate portfolio sorts that control for momentum, reversal, and an ML model based only on momentum and reversal.

Economic message: MLER contains known components, but a large part of its predictive power is distinct from the classic momentum and reversal effects.

Table 11

Relations between ML-Based Forecasts, Momentum, and Reversal. This table presents the results of Fama and MacBeth (1973) regression analyses examining the ability of Mom, Rev, and $MLER^{Mom,Rev}$ to explain variation in $MLER$. The process for conducting the Fama and MacBeth (1973) regression analyses is exactly as described in Table 10. The dependent variable in the regressions is $MLER$ and the independent variables are combinations of Mom, Rev, and $MLER^{Mom,Rev}$. Results for different regression specifications are shown in different columns of the table. The table presents the time-series averages of monthly cross-sectional regression coefficients, along with t-statistics, calculated following Newey and West (1987) using 12 lags, testing the null hypothesis that the average coefficient is equal to zero (in parentheses). The rows labeled "Mom", "Rev", and "$MLER^{Mom,Rev}$" present results pertaining to the coefficient on the indicated variable. The row labeled "Adj. R^2" shows the average adjusted R^2 from the cross-sectional regressions. The analyses cover sample months t from July 1963 through December 2022, inclusive.

Table 11: relations with momentum and reversal

Table 12

Multivariate Portfolio Analysis - Control for Momentum and Reversal. This table presents the results of univariate, bivariate, and trivariate portfolio analyses examining the ability of the ML-based forecasts to predict the cross-section of future stock returns after controlling for momentum and reversal. The procedure used to generate the univariate portfolios is identical to that used to generate the portfolios in Table 3, except we create five quintile portfolios instead of 10 decile portfolios. The bivariate portfolios are constructed by sorting all stocks into quintiles based on a control variable, either Mom, Rev, or $MLER^{Mom,Rev}$, and then within each control variable quintile, into five $MLER$ portfolios. The trivariate portfolios are formed by independently sorting stocks into quintiles of Mom and Rev, and then sorting stocks in each of the 25 groups formed by the intersections of the Mom and Rev quintiles, into $MLER$ quintiles. Breakpoints for all sorts are calculated using only NYSE-listed stocks. Values of $MLER$, Mom, Rev, and $MLER^{Mom,Rev}$ are calculated as of the end of month $t-1$. The month t excess return of each of the resulting portfolios is taken to be the market capitalization-weighted average month t excess return of all stocks in the given portfolio, with market capitalization calculated as of the end of month $t-1$. For the bivariate (trivariate) portfolios, the month t excess return of the $MLER$ quintile k portfolio is taken to be the equal-weighted average, across all quintiles of the control variable (across all 25 Mom and Rev groups), of the $MLER$ quintile k portfolio. Finally, the $MLER$ $5-1$ portfolio excess return is taken to be the difference between the $MLER$ quintile five and quintile one portfolio excess returns. The table presents the time-series averages of the monthly portfolio excess returns for each of the $MLER$ quintile portfolios. The column labeled "Control Variable(s)" indicates the control variable(s) used to construct the portfolios. The columns labeled "$MLER$ 1" through "$MLER$ 5" and "$MLER$ $5-1$" present the average monthly excess portfolio returns along with t-statistics, calculated following Newey and West (1987) using 12 lags, testing the null hypothesis that the average monthly excess return of the given portfolio is equal to zero. All excess returns are reported in percent per month. The analyses cover return months t from July 1963 through December 2022, inclusive.

Control Variable(s)	$MLER$ 1	$MLER$ 2	$MLER$ 3	$MLER$ 4	$MLER$ 5	$MLER$ $5-1$

Table 12: portfolio controls for momentum and reversal

5.7 Tables 13–16: image-based ML forecasts and technical-signal controls

Read:

- Table 13 examines the relation between MLER and the image-based forecasts of Jiang, Kelly, and Xiu.
- Tables 14–15 show that MLER remains powerful even in the 2001–2020 subperiod and after controlling for image-based forecast factors.
- Table 16 shows that MLER remains predictive after controlling for technical signals and trading-friction variables.

Economic message: The machine charting signal is not redundant with earlier image-based ML signals, technical trading rules, or market-friction predictors.

Table 13

Relations between ML-Based Forecasts and Image-Based Forecasts. This table presents the results of Fama and MacBeth (1973) regression analyses examining the ability of $I60/R20$, $I20/R20$, and $I5/R20$ to explain variation in $MLER$. The process for conducting the Fama and MacBeth (1973) regression analyses is exactly as described in Table 10. The dependent variable in the regressions is $MLER$ and the independent variables are combinations of $I60/R20$, $I20/R20$, and $I5/R20$. Results for different regression specifications are shown in different columns of the table. The table presents the time-series averages of monthly cross-sectional regression coefficients, along with t-statistics, calculated following Newey and West (1987) using 12 lags, testing the null hypothesis that the average coefficient is equal to zero (in parentheses). The rows labeled "$I60/R20$", "$I20/R20$", and "$I5/R20$" present results pertaining to the coefficient on the indicated variable. The row labeled "Adj. R^2" shows the average adjusted R^2

Table 13: relation to image-based forecasts

Table 14

Portfolio Analysis - 200102-202001. This table presents the results of a portfolio analysis examining the ability of $MLER$ to predict the cross-section of future stock returns during the 200102-202001 period. The analysis is identical to that whose results are shown in Table 3 except that the results presented in this table are for return months t from February 2001 through January 2020, inclusive.

Value	$MLER$ 1	$MLER$ 2	$MLER$ 3	$MLER$ 4	$MLER$ 5	$MLER$ 6	$MLER$ 7	$MLER$ 8	$MLER$ 9	$MLER$ 10	$MLER$ $10-1$
$\bar{r}$	−0.02	0.41	0.50	0.56	0.52	0.71	0.68	0.58	0.68	0.80	0.82
	(−0.03)	(0.86)	(1.22)	(1.46)	(1.56)	(2.58)	(2.37)	(1.90)	(2.36)	(2.32)	(1.72)
SD	7.52	6.21	5.57	5.30	4.44	4.22	4.00	3.98	3.88	4.50	5.75
Sharpe	−0.01	0.23	0.31	0.36	0.40	0.59	0.59	0.51	0.61	0.62	0.49

Table 14: 2001–2020 analysis

Table 15

Alphas Using Image-Based Forecast Factors. This table presents the results of factor model regressions of the excess returns of the $MLER$ $10-1$ portfolio on the excess returns of factors in different factor models. The $MLER$ $10-1$ portfolio is the same as that whose average excess returns are shown in Table 3. The excess return of the F_X factor, for $X \in \{I60/R20, I20/R20, I5/R20\}$, is that of the long-short portfolio formed by sorting on both market capitalization and X, constructed using a methodology, described in Section 5.4.2, similar to that of Fama and French (1993). The column labeled "Model" indicates the factor model, where "Base Model" refers to the model indicated in the header of columns other than the "Model" and "Value" columns and "Base Model+ $F_{I60/R20} + F_{I20/R20} + F_{I5/R20}$" refers to a model that augments the factors in the model shown in the column header with $F_{I60/R20}$, $F_{I20,R20}$, and $F_{I5/R20}$. The rows with "α" and "Adj. R^2" in the "Value" column present the intercept and adjusted R^2, respectively, of the regression. The rows with "β_{F_X}" in the "Value" column present the slope coefficient on F_X from the regression. Alphas are reported in percent per month. The values in parentheses are t-statistics, calculated following Newey and West (1987) using 12 lags, testing the null hypothesis that the average monthly alpha or slope coefficient is equal to zero. The analyses cover return months t from February 2001 through January 2020, inclusive.

Model	Value	CAPM	FF	FFC	FFC+REV	FF5	Q
Base Model	α	1.15	1.19	0.75	0.64	0.65	0.45
		(2.50)	(2.56)	(2.38)	(2.48)	(1.88)	(1.16)
	Adj. R^2	19.88%	22.42%	57.20%	65.38%	30.82%	48.96%
Base Model+ $F_{I60/R20} + F_{I20/R20} + F_{I5/R20}$	α	1.01	1.10	0.67	0.61	0.72	0.47
		(2.79)	(3.09)	(2.30)	(2.44)	(2.64)	(1.36)
	$\beta_{F_{I60/R20}}$	1.20	1.20	0.18	0.31	0.83	0.38
		(3.29)	(3.31)	(0.67)	(1.41)	(1.83)	(1.39)
	$\beta_{F_{I20/R20}}$	−0.33	−0.39	0.18	0.19	−0.31	−0.11
		(−1.10)	(−1.32)	(0.70)	(0.73)	(−1.08)	(−0.42)
	$\beta_{F_{I5/R20}}$	−0.23	−0.28	−0.12	−0.28	−0.22	−0.23
		(−0.78)	(−0.93)	(−0.51)	(−1.47)	(−0.68)	(−0.98)
	Adj. R^2	27.80%	29.82%	56.94%	65.61%	33.70%	50.70%

Table 15: image-based forecast factors

Table 16

Multivariate Portfolio Analysis - Control for Technical Signals. This table presents the results of bivariate portfolio analyses examining the ability of the ML-based forecasts to predict the cross-section of future stock returns after controlling for other technical signals. The technical signals are those used by Neely et al. (2014, NRTZ) and Freyberger et al. (2020, FNW), described in Section 5.5. Panel A presents the average excess returns of $MLER$ 10 − 1 portfolios designed to be neutral to different NRTZ technical signals (indicated in the column headers). At the end of each month $t-1$, stocks with each value of the given technical signal (either 0 or 1), are sorted into decile portfolios based on an ascending ordering of $MLER$ using breakpoints calculated from NYSE-listed stocks. The month t excess return of each portfolio is then taken to be the market capitalization-weighted average month t excess return of all stocks in the portfolio, with market capitalization calculated as of the end of month $t-1$. The $MLER$ 10 − 1 portfolio excess return is taken to be the equal-weighted average excess return of the two decile 10 $MLER$ portfolios minus that of the two decile 1 portfolios. Panel B presents the average excess returns of $MLER$ 5 − 1 portfolios designed to be neutral to different FNW technical signals (indicated in the column headers). The methodology used to construct these $MLER$ 5 − 1 portfolios is identical to that used to construct the bivariate portfolios examined in Table 12 except that one of the FNW technical signals is used as the first sort variable. The values in parentheses are t-statistics, calculated following Newey and West (1987) using 12 lags, testing the null hypothesis that the average monthly excess return is equal to zero. The analyses cover return months t from July 1963 through December 2022, inclusive.

Panel A: $MLER$ 10 − 1 Port. Avg. Excess Return - Control for NRTZ Variables

$I_{MA(1)\geq MA(9)}$	$I_{MA(1)\geq MA(12)}$	$I_{MA(2)\geq MA(9)}$	$I_{MA(2)\geq MA(12)}$	$I_{MA(3)\geq MA(9)}$	$I_{MA(3)\geq MA(12)}$	$I_{P_t\geq P_{t-9}}$	$I_{P_t\geq P_{t-12}}$	$I_{OBV MA(1)\geq OBV MA(9)}$	$I_{OBV MA(1)\geq OBV MA(12)}$	$I_{OBV MA(2)\geq OBV MA(9)}$	$I_{OBV MA(2)\geq OBV MA(12)}$	$I_{OBV MA(3)\geq OBV MA(9)}$	$I_{OBV MA(3)\geq OBV MA(12)}$
1.13	1.19	1.15	1.18	1.07	1.08	1.15	1.10	1.17	1.15	1.15	1.11	1.10	1.09
(5.67)	(7.01)	(6.70)	(7.63)	(6.50)	(6.68)	(6.73)	(6.77)	(6.07)	(6.06)	(6.10)	(6.03)	(5.77)	(6.07)

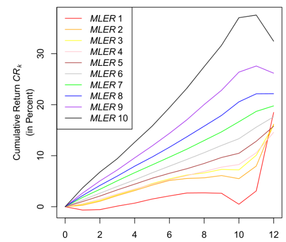
Panel B: $MLER$ 5 − 1 Port. Avg. Excess Return - Control for FNW Variables

$TotVol$	$IdioVol$	$Price/High$	Max	σ_{Turn}	σ_{Volume}	β_{FP}	β_{LN}	DTO	SUV	$Turn$	AT	$MktCap$	$MktCap_{IndAdj}$
0.75	0.83	0.75	0.80	0.83	0.94	0.70	0.76	0.80	0.79	0.81	0.97	1.02	0.93
(6.85)	(6.88)	(4.66)	(6.87)	(6.17)	(5.76)	(6.26)	(6.59)	(6.02)	(5.11)	(6.16)	(6.55)	(6.20)	(5.60)

g the adaptive group LASSO methodology to 62 stock-level char-
cs, which include the technical variables mentioned above, past
ased variables including Mom and Rev, and many variables
cted from financial statement data. The results of these tests,
d in detail and presented in Section XXIV and Table A21 of the
Appendix, show that the $MLER$ 10−1 portfolio generates pos-
onomically large, and highly statistically significant alpha with
to traditional factor models augmented with the excess returns
's long-short portfolio.
conclude from these tests that the predictive power of the ML-
recasts is not subsumed by previously-studied technical signals.

ual analysis of charts

final analyses characterizing the predictive power of $MLER$
lescribe what charts that predict high or low future returns look
begin by plotting the average chart for the focal $MLER$ decile
os that we examined in Table 3. Specifically, for each month t

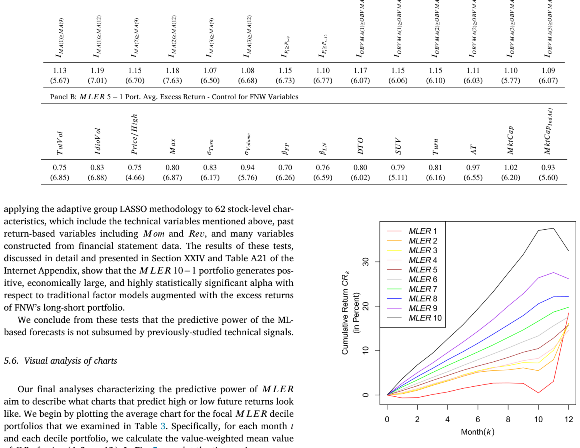


5.8 Figures 5–7 and Table 17: chart interpretation and optimization validation

Read:

- Figure 5 plots average charts across MLER decile portfolios.
- Figure 6 shows average charts for MLER quintiles after removing much of the momentum and reversal effect.
- Table 17 evaluates many alternative ML models out of sample.
- Figure 7 shows that optimization-period performance is positively related to test-period performance.

Economic message: The visual charts illustrate that the ML model extracts subtle shape information, and the optimization validation shows that model selection using the early period is meaningful.



$I_{MA(1)\geq MA(9)}$	$I_{MA(1)\geq MA(12)}$	$I_{MA(2)\geq MA(9)}$	$I_{MA(2)\geq MA(12)}$	$I_{MA(3)\geq MA(9)}$	$I_{MA(3)\geq MA(12)}$	$I_{P_t\geq P_{t-9}}$	$I_{P_t\geq P_{t-12}}$	$I_{OBVMA(1)\geq OBVMA(9)}$	$I_{OBVMA(1)\geq OBVMA(12)}$	$I_{OBVMA(2)\geq OBVMA(9)}$	$I_{OBVMA(2)\geq OBVMA(12)}$	$I_{OBVMA(3)\geq OBVMA(9)}$	$I_{OBVMA(3)\geq OBVMA(12)}$
1.13 (5.67)	1.19 (7.01)	1.15 (6.70)	1.18 (7.63)	1.07 (6.50)	1.08 (6.68)	1.15 (6.73)	1.10 (6.77)	1.17 (6.07)	1.15 (6.06)	1.15 (6.10)	1.11 (6.03)	1.10 (5.77)	1.09 (6.07)

Panel B: $MLER$ 5 − 1 Port. Avg. Excess Return - Control for FNW Variables

$TotVol$	$IdioVol$	$Price/High$	Max	σ_{Turn}	σ_{Volume}	β_{FP}	β_{LN}	DTO	SUV	$Turn$	AT	$MktCap$	$MktCap_{IndAdj}$
0.75 (6.85)	0.83 (6.88)	0.75 (4.66)	0.80 (6.87)	0.83 (6.17)	0.94 (5.76)	0.70 (6.26)	0.76 (6.59)	0.80 (6.02)	0.79 (5.11)	0.81 (6.16)	0.97 (6.55)	1.02 (6.20)	0.93 (5.60)

applying the adaptive group LASSO methodology to 62 stock-level characteristics, which include the technical variables mentioned above, past return-based variables including Mom and Rev, and many variables constructed from financial statement data. The results of these tests, discussed in detail and presented in Section XXIV and Table A21 of the Internet Appendix, show that the $MLER$ 10 − 1 portfolio generates positive, economically large, and highly statistically significant alpha with respect to traditional factor models augmented with the excess returns of FNW's long-short portfolio.

We conclude from these tests that the predictive power of the ML-based forecasts is not subsumed by previously-studied technical signals.

5.6. Visual analysis of charts

Our final analyses characterizing the predictive power of $MLER$ aim to describe what charts that predict high or low future returns look like. We begin by plotting the average chart for the focal $MLER$ decile portfolios that we examined in Table 3. Specifically, for each month t and each decile portfolio, we calculate the value-weighted mean value

Figure 5: average charts for MLER *deciles*

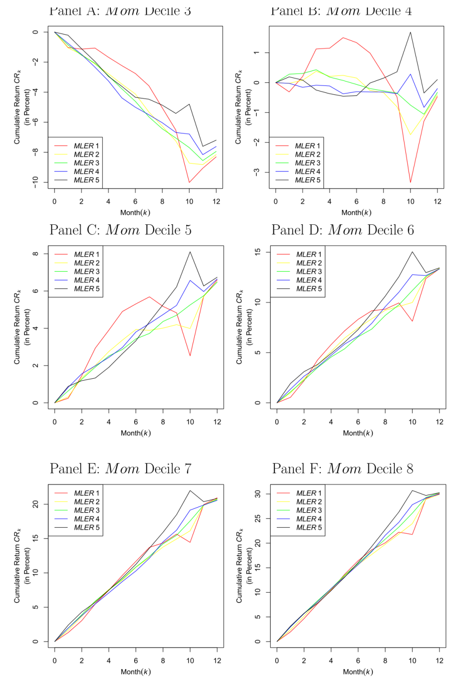


Figure 6: average charts excluding momentum/reversal

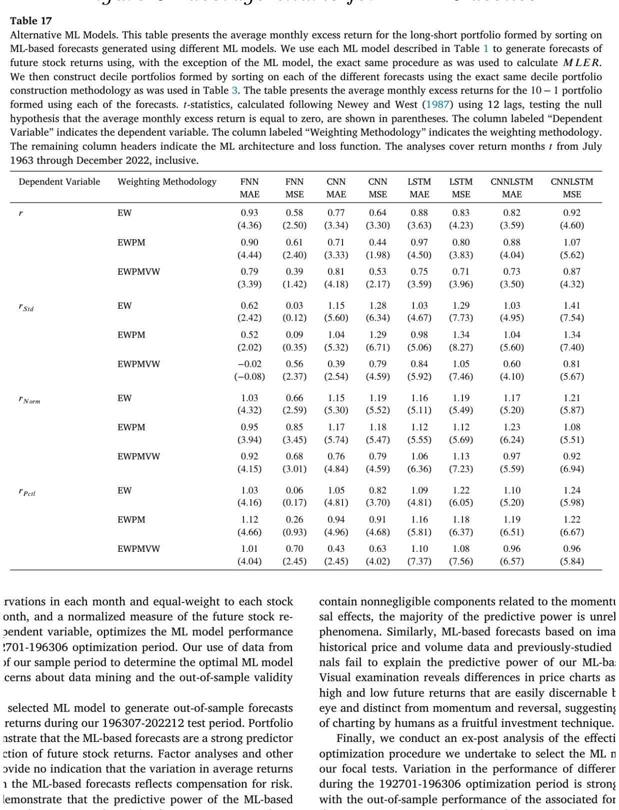


Table 17

Alternative ML Models. This table presents the average monthly excess return for the long-short portfolio formed by sorting on ML-based forecasts generated using different ML models. We use each ML model described in Table 1 to generate forecasts of future stock returns using, with the exception of the ML model, the exact same procedure as was used to calculate $MLER$. We then construct decile portfolios formed by sorting on each of the different forecasts using the exact same decile portfolio construction methodology as was used in Table 3. The table presents the average monthly excess returns for the 10 − 1 portfolio formed using each of the forecasts. t-statistics, calculated following Newey and West (1987) using 12 lags, testing the null hypothesis that the average monthly excess return is equal to zero, are shown in parentheses. The column labeled "Dependent Variable" indicates the dependent variable. The column labeled "Weighting Methodology" indicates the weighting methodology. The remaining column headers indicate the ML architecture and loss function. The analyses cover return months t from July 1963 through December 2022, inclusive.

Dependent Variable	Weighting Methodology	FNN MAE	FNN MSE	CNN MAE	CNN MSE	LSTM MAE	LSTM MSE	CNNLSTM MAE	CNNLSTM MSE
r	EW	0.93 (4.36)	0.58 (2.50)	0.77 (3.34)	0.64 (3.30)	0.88 (3.63)	0.83 (4.23)	0.82 (3.59)	0.92 (4.60)
	EWPM	0.90 (4.44)	0.61 (2.40)	0.71 (3.33)	0.44 (1.98)	0.97 (4.50)	0.80 (3.83)	0.88 (4.04)	1.07 (5.62)
	EWPMVW	0.79 (3.39)	0.39 (1.42)	0.81 (4.18)	0.53 (2.17)	0.75 (3.59)	0.71 (3.96)	0.73 (3.50)	0.87 (4.32)
r_{Std}	EW	0.62 (2.42)	0.03 (0.12)	1.15 (5.60)	1.28 (6.34)	1.03 (4.67)	1.29 (7.73)	1.03 (4.95)	1.41 (7.54)
	EWPM	0.52 (2.02)	0.09 (0.35)	1.04 (5.32)	1.29 (6.71)	0.98 (5.06)	1.34 (8.27)	1.04 (5.60)	1.34 (7.40)
	EWPMVW	−0.02 (−0.08)	0.56 (2.37)	0.39 (2.54)	0.79 (4.59)	0.84 (5.92)	1.05 (7.46)	0.60 (4.10)	0.81 (5.67)
r_{Norm}	EW	1.03 (4.32)	0.66 (2.59)	1.15 (5.30)	1.19 (5.52)	1.16 (5.11)	1.19 (5.49)	1.17 (5.20)	1.21 (5.87)
	EWPM	0.95 (3.94)	0.85 (3.45)	1.17 (5.74)	1.18 (5.47)	1.12 (5.55)	1.12 (5.69)	1.23 (6.24)	1.08 (5.51)
	EWPMVW	0.92 (4.15)	0.68 (3.01)	0.76 (4.84)	0.79 (4.59)	1.06 (6.36)	1.13 (7.23)	0.97 (5.59)	0.92 (6.94)
r_{Pctl}	EW	1.03 (4.16)	0.06 (0.17)	1.05 (4.81)	0.82 (3.70)	1.09 (4.81)	1.22 (6.05)	1.10 (5.20)	1.24 (5.98)
	EWPM	1.12 (4.66)	0.26 (0.93)	0.94 (4.96)	0.91 (4.68)	1.16 (5.81)	1.18 (6.37)	1.19 (6.51)	1.22 (6.67)
	EWPMVW	1.01 (4.04)	0.70 (2.45)	0.43 (2.45)	0.63 (4.02)	1.10 (7.37)	1.08 (7.56)	0.96 (6.57)	0.96 (5.84)

rvations in each month and equal-weight to each stock onth, and a normalized measure of the future stock re- pendent variable, optimizes the ML model performance 2701-196306 optimization period. Our use of data from of our sample period to determine the optimal ML model cerns about data mining and the out-of-sample validity

selected ML model to generate out-of-sample forecasts returns during our 196307-202212 test period. Portfolio nstrate that the ML-based forecasts are a strong predictor ction of future stock returns. Factor analyses and other ovide no indication that the variation in average returns n the ML-based forecasts reflects compensation for risk. lemonstrate that the predictive power of the ML-based

contain nonnegligible components related to the momentu sal effects, the majority of the predictive power is unrel phenomena. Similarly, ML-based forecasts based on ima historical price and volume data and previously-studied nals fail to explain the predictive power of our ML-ba Visual examination reveals differences in price charts as high and low future returns that are easily discernable b eye and distinct from momentum and reversal, suggesting of charting by humans as a fruitful investment technique.

Finally, we conduct an ex-post analysis of the effecti optimization procedure we undertake to select the ML n our focal tests. Variation in the performance of differer during the 192701-196306 optimization period is strong with the out-of-sample performance of the associated for

Table 17: alternative ML models

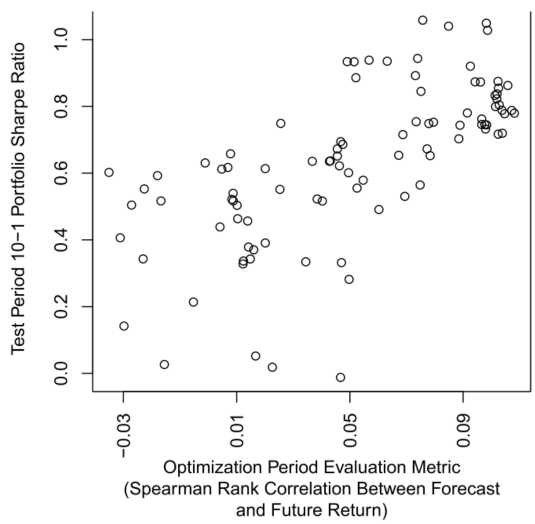


Figure 7: optimization versus test performance

6 Short summary

- The paper tests weak-form market efficiency using ML forecasts based only on past return charts.
- The input vector is twelve monthly cumulative returns from a one-year price plot.

- The model is selected using only 1927–1963 data; the main tests use 1963–2022 data.
- The optimized model is CNNLSTM with MSE loss, equal weighting per month and per stock, and normalized returns.
- MLER decile portfolios produce a 10-minus-1 return of about 1.08% per month.
- Factor-model alphas are large and statistically significant.
- The result is robust across most subperiods and remains strong among large stocks.
- Forecasting functions estimated in different windows are highly related and produce similar long-short trades.
- Partial-dependence and Fama–MacBeth analyses show nonlinearities and interactions.
- Momentum and reversal explain part, but not all, of the signal.
- Image-based ML forecasts, technical signals, and trading-friction controls do not subsume MLER.
- Models that perform well during optimization also perform well during the out-of-sample test period.
- The strongest interpretation is not that every historical chart pattern is profitable, but that a flexible and disciplined mapping from chart shape to return ranks contains economically meaningful information.
- The paper's evidence is especially challenging because it remains strong after risk adjustment, after large-stock restrictions, after recent-subperiod tests, and after controls for other technical predictors.

7 Conclusion

7.1 Core conclusion

Murray, Xia, and Xiao show that machine learning can extract strong predictive information from historical return charts. Their MLER signal predicts the cross-section of future stock returns with economically large and statistically significant returns, even though it uses only past price-performance data.

7.2 Economic intuition

The weak-form EMH says that price charts should not contain exploitable return-predictive information. The paper argues that earlier technical-analysis research may have failed because hand-crafted rules are too simple. Neural networks can learn nonlinear and interactive chart patterns that humans or simple rules may miss.

7.3 Takeaway

Technical analysis has merit when implemented as machine charting rather than fixed chart rules. The evidence does not merely rediscover momentum and reversal; it reveals more complex shape-based return predictability. The paper therefore challenges the weak form of market efficiency and shows that disciplined ML model selection can deliver genuine out-of-sample performance.